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Pick and mix reliable pseudo labels for scribble-supervised medical image segmentation

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ABSTRACT

Scribble-supervised segmentation methods have attracted significant attention in the field of medical imaging because of their potential to alleviate the data annotation burden. However, these methods often underperform due to a lack of sufficient supervision. Various methods have attempted to enrich the supervisory signals in different ways, including mixing pseudo labels from different samples (referred as Mixup-based method). However, these methods primarily focus on the quantity of enriched supervisory signals, disregarding their quality. This oversight presents a major drawback in that low-quality signals are often contaminated with the noise, thus can lead to undermine performance. Therefore, it is crucial to not only introduce diverse supervisory signals but also ensure their quality and reliability. Motivated by this understanding, we propose a new framework named Pick & Mix, which builds upon the Mixup-based method. In the first step, we leverage the consistency of intra-class features to assess the reliability of pseudo-labels. To enhance the quality of pseudo labels, we assign lower weights to those unreliable pseudo-labels to mitigate the noise effect in the training process. Furthermore, we utilize a threshold to pick reliable pseudo-labels based on their reliability score. In the second step, we mix the reliable pseudo-labels from various samples and generate corresponding mixed images to provide richer supervisory signals for model training. In this manner, we enhance the quality of supervisory signals by generating and picking reliable ones, as well as enrich the quantity of these signals through a process of mixing. Finally, we evaluated our framework on three publicly available datasets: ACDC, MSCMRseg, and BraTS2020. The experimental results demonstrate that our approach achieves state-of-the-art performance.

1. Introduction

Medical image segmentation plays a pivotal role in the field of medical image analysis. In recent years, Convolutional Neural Networks (CNNs) [1,2] and Transformers [3] have made significant progress in automatic medical image segmentation [4–6]. However, most of existing methods require large-scale images with precise pixel-level dense annotation for training. The acquisition of such a dataset is not only costly but also time-consuming [7,8]. To ease the challenges associated with data annotation, a variety of methods have been proposed [9–11]. Among these approaches, methods utilizing scribble supervision have demonstrated considerable potential. As shown in Fig. 1(a), scribble-supervised semantic segmentation leverages some rough lines to train the segmentation model [12]. However, the performance of the segmentation is restricted due to the inadequacy of the annotations.

To address this issue, a straightforward yet potent strategy involves enhancing the supervisory signal. Underpinned by this concept, current scribble-supervised studies can be categorized into three distinct groups. The first category utilizes Generative Adversarial Networks (GANs) [13] to enhance the supervisory signal by acquiring shape knowledge from additional unpaired fully-annotated masks [14,15]. The second category, referred as Prior-based methods, leverages prior knowledge to expand scribble annotations [16,17]. The third category, termed as Mixup-based methods, generates supplementary supervisory signals by blending pseudo labels from various samples [18,19]. However, these methods primarily focus on introducing new supervisory signals, neglecting the assessment of the quality of these newly incorporated signals. The introducing of supervisory signals with lower quality can potentially misguide the model, thereby limiting its performance.

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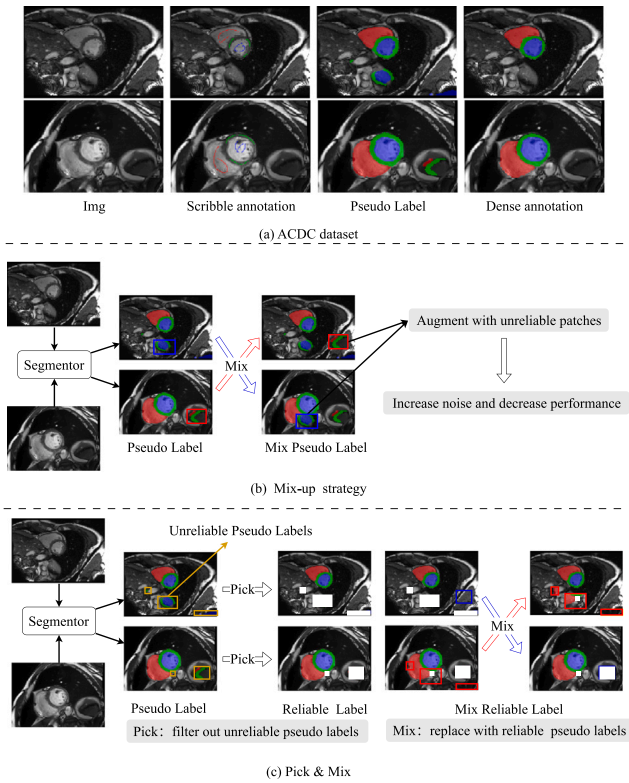

Fig. 1. (a): Examples from the ACDC dataset. Due to the limited quantity of supervisory signals, the model tends to produce predictions of lower quality. (b): Simply employ a mix-up strategy on pseudo labels could potentially exacerbate the amplification of noisy information. (c): Two phases of our proposed Pick & Mix framework. We pick reliable pseudo labels to enhance the quality of supervisory signals and mix them to increase the quantity.

For instance, as depicted in Fig. 1(b), the implementation of mix-up strategies enhances the volume of supervised signals. However, it simultaneously infuses additional noise, thereby constraining the performance of the model. This highlights the need for strategies that not only introduce new supervisory signals but also ensure their quality to prevent the model from being misled.

In this paper, following the Mixup-based methods, we propose a novel framework, termed “Pick & Mix”. Pick & Mix filters out unreliable pseudo labels to enhance the quality of supervisory signals and mixes reliable ones to increase their quantity for scribble-supervised medical image segmentation. As illustrated in Fig. 1(c), Pick & Mix contains two main steps. In the initial step, we assess the reliability of various pseudo-labels by leveraging the consistency of intra-class features. Subsequently, with the objective of procuring high-quality pseudo-labels, we adopt a two-pronged approach: firstly, we assign dynamic weights to pseudo-labels based on their reliability. Lower weights are allocated to pseudo-labels with lower reliability to mitigate the noise effect, thereby enabling the model to generate highly reliable pseudo-labels. Secondly, we employ a threshold-based selection process to choose pseudo-labels with high reliability. In the second step, we further augment the quantity of supervised signals by bi-directionally mixing reliable pseudo-labels from various samples. The unreliable pseudo-labels are subsequently replaced with trustworthy labels from corresponding spatial locations in other samples, leading to the creation of a novel mixed reliable labels and correspond mixed image. Through the use of Pick & Mix, we are able to amplify both the quality and quantity of the supervisory signals for the model training. Benefited from the proposed framework, we achieved the state-of-the-art (SOTA) on three public datasets: ACDC [20], MSCMRseg [21,22] and BraTS2020 [23].

To summarize, the contributions of this work are three-fold:

- We propose a balanced strategy for scribble-supervised medical image segmentation tasks, which not only advocates for the augmentation of the quantity of supervisory signals, but also emphasizes the imperative of ensuring their quality.
- We introduce a framework named as Pick & Mix, which bi-directionally incorporates reliable pseudo labels from a variety of samples to deliver a precise augmented supervision signal. This approach mitigates the issue of noise during model training and enhances the quality of the supervised signal.
- We have conducted extensive experiment on ACDC, MSCMRseg and BraTS22020 datasets. The comparison and ablation studies demonstrate that our method outperforms existing SOTA scribble-supervised medical image segmentation methods, as well as the effectiveness of each component.

2. Related works

2.1. Scribble-supervised medical image segmentation

In the method of scribble-based segmentation, annotations are provided using only a minimal fraction of pixels. This method is particularly beneficial in scenarios where comprehensive pixel-level annotations are either impractical or excessively costly. The crux to address the problem is how to get more additional supervised information to train the model. Previous works can be broadly classified into three categorizes:(1) GAN-based methods. (2) Prior-based methods. (3) Mixup-based methods. The GAN-based methods learns the shape knowledge from unpaired mask with GAN network. For example, Valvano et al. utilized unpaired segmentation mask to train a multi scale GAN to generate realistic segmentation masks at multiple resolutions [15]. However, additional resources are required. Prior-based method expanded scribble annotations by incorporate prior knowledge from various ways [16,17]. For example, Zhou et al. [17] combined superpixel-guided scribble walking with Class-wise contrastive regularization to enhance the performance of model. Li et al. [16] utilized both vision and class embeddings to enhance their model, leveraging multimodal information in the process. The Mixup-based method mixed the pseudo labels from various samples to generate supplementary supervisory signals [18,19]. Luo et al. [19] introduced a dynamic approach to mix pseudo labels provided from two branches, aiming to offer a more robust supervision signal. Meanwhile, Zhang and Zhuang [18] proposed a method involving global and local consistency loss to regulate the model trained with mixed pseudo labels.

Previous methods have enhanced the model performance by generating additional supervisory signals. Furthermore, we posit that the supervisory signals introduced should undergo a screening process to prevent the model from being misled by erroneous signals. Consequently, we propose a method to selectively pick and mix reliable pseudo labels with the aim of enhancing the segmentation performance of model.

2.2. Mixup augmentations

Data augmentation is a crucial strategy in enhancing the generalization ability of neural networks and preventing overfitting to limited training data [24,25]. A subset of these strategies, known as Mixup augmentations, involve the combination of two images and their corresponding labels [26–31]. These methods offer an advantage over traditional augmentation techniques such as rotation and flipping, as they can increase the scribble annotations of an augmented image through a mix operation. For instance, MixUp [26] performs linear interpolation between two images and their labels. Furthermore, existing works extended MixUp operations from input images to hidden features. CutOut [27] randomly omits square regions of images. Based

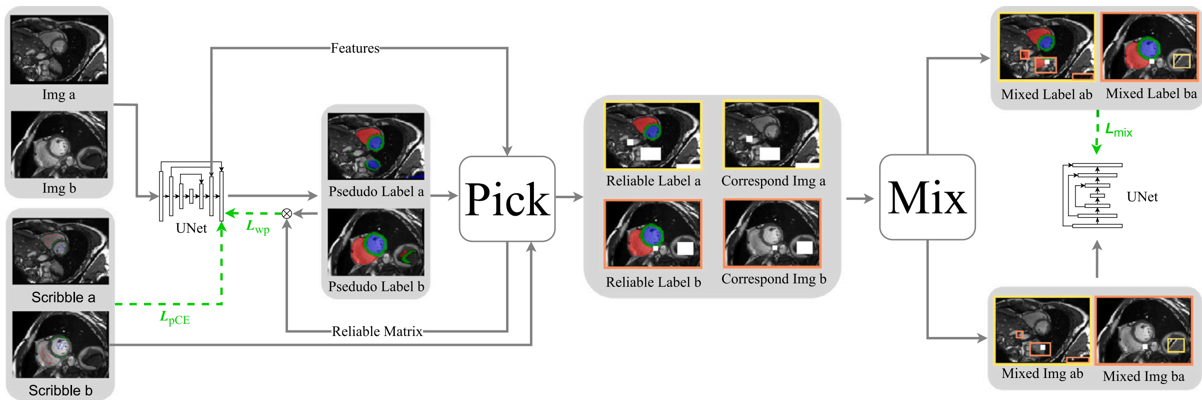

Fig. 2. The pipeline of Pick & Mix. Indeed, the two UNets depicted in the figure represent the same network. The training of our model is end-to-end. The loss functions are annotated with green. The hollow area observed in the mixed images and mixed labels, indicating that the predictions for this region are unreliable in both samples.

on it, CutMix [28] replaces these omitted areas with patches from other images. Co-mixup [29] extended the mixup operation from two images to multiple images, promoting the diversity of mixed images. Puzzle Mix [30] introduced a new mixup method based on saliency and local statistics. While the images produced through mixup methods may not appear realistic, the mixed soft labels they generate can provide additional information that facilitates the training of models [31]. In this paper, we employ selective mixing techniques to generate reliable supervisory signals for enhancing the model's performance under scribble supervision.

3. Method

In this section, we initially outline the process of learning with scribble annotation in Section 3.1, which serves as the foundation for our model. After that, two main steps of our method are described in Sections 3.2 and 3.3. Finally, we describe the training losses in our proposed Pick & Mix in Section 3.4.

3.1. Learning with scribble annotations

Given an input image $X \in \mathbb{R}^{H\times W}$ with sparse annotations $S \in \mathbb{R}^{H\times W}$. The predictions are denoted as $Y \in \mathbb{R}^{N\times H\times W}$. The N denotes number of class and H, W represent the size of height and width. Here we directly utilize spare annotation S for supervision with a partial cross-entropy loss L_{pCE} as follows:

$$L_{pCE}(y, s) = -\sum_{n\in N}\sum_{i\in\omega_s} s_n^i log y_n^i, \tag{1}$$

where ω_s is the set of labeled pixels in S. y_n^i is the predicted probability of pixel i belong to class n. s_n^i represents the one-hot scribble annotations, indicating whether the ith pixel is a member of the nth class.

Inadequate supervisory information does not ensure that the model generates high-quality predictions. Hence, we introduce a novel framework in this study, termed Pick & Mix, designed to enhance both the quality and quantity of supervisory signals, respectively. As shown in Fig. 2, our approach contains two primary steps: (1) Pick Reliable Pseudo Labels. (2) Mix Reliable Pseudo Labels. We will cover these two steps in detail in subsequent sections.

3.2. Pick reliable pseudo labels

The target of the first step "Pick" is to enhance the quality of supervisory signals. To this end, we initially focus on enhancing the reliability of the pseudo-label. Subsequently, we implement a threshold mechanism to select those pseudo-labels that exhibit a higher degree of reliability.

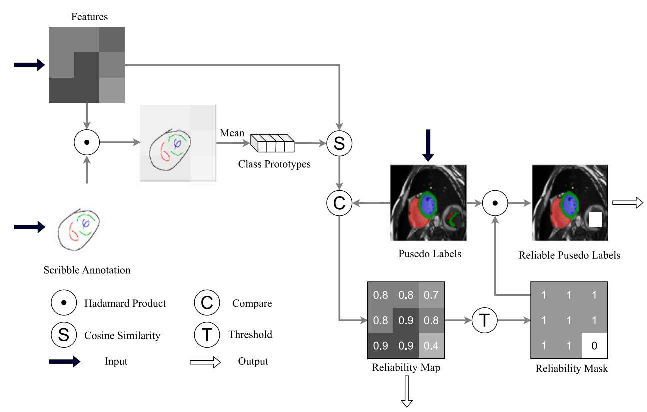

Fig. 3. The pipeline of pick reliable pseudo labels.

Primarily, it is essential to assess the reliability of pseudo-labels. Inspired by the intra-class feature consistency, we employ class prototypes as a tool for assessing the reliability of these pseudo-labels. As shown in Fig. 3, we first employ scribble-annotations to generate a mask to identify the features associated with a specific class, is computed by:

$$mask_n = \mathbb{I}(S = n), \tag{2}$$

where $\mathbb{I}$ represents the indicator function, which $mask_n = 1$ when the scribble annotation of correspond pixel is class n. After that, we derive the prototype for the nth class, denoted as $CP_n \in \mathbb{R}^{c\times 1\times 1}$, by computing the average of all features that are labeled as the nth class. Note that we extract the features $F \in \mathbb{R}^{C\times H\times W}$ from penultimate layer, which have the same spatial size of scribble-annotations. The C denotes channel size. This process can be denoted as:

$$CP_n = (F * mask_n)/sum(mask_n), \tag{3}$$

Once the class prototype is extracted, we proceed to compute the pixel-wise cosine similarity sim_n between the CP_n and the F as:

$$sim_n = cos(F, CP_n) = \frac{F^T CP_n}{\|F^T\| \|CP_n\|}. \tag{4}$$

After that, we calculate the distance between the sim_n and prediction y_n to determine the reliability of nth class pseudo labels, which is denoted as:

$$R_n = 1 - |sim_n - y_n|. \tag{5}$$

Indeed, a lower value in this context signifies lower reliability. Based on the category values of the pseudo-labels $P = argmax(Y)$, we obtained the corresponding reliability r from R_n, which can be denoted as:

$$r = \sum_{n=1}^{N} R_n \cdot \mathbb{I}(P = n), \tag{6}$$

Finally, we utilize the r as a pixel-wise weight for cross entropy loss to help the model learn from pseudo labels. The loss can be denoted as:

$$L_{wp} = -\frac{1}{H \times W \times N} \sum_{i}^{H \times W} \sum_{n=1}^{N} r^i \cdot \mathbb{I}(P^i = n) \cdot \log(y_n^i). \tag{7}$$

By placing less weight on unreliable pseudo-labels, the model can avoid being misled by the noisy information and produce high-quality pseudo labels.

Additionally, we utilize a threshold to generate reliability mask RM based on the reliability map as:

$$RM = r > T, \tag{8}$$

where r is extracted from R_n based on the category values of the pseudo-labels P at the corresponding spatial locations. And the T denote threshold, we set it as 0.9 in our approach. Finally, we obtain the reliable pseudo-label by performing the Hadamard product between RM and the pseudo-labels P.

3.3. Mix reliable pseudo labels

In this step, we employ a technique known as Mix-up to augment the quantity of our supervised signals. Contrary to previous methodologies, our approach applies the Mix-up technique exclusively to reliable pseudo-labels. This is because indiscriminate application of the mixup strategy could potentially replace reliable pseudo-labels with less reliable ones, thereby amplifying the noise and degrading the quality of predictions. Therefore, we propose that careful selection of reliable pseudo-labels for hybrid is crucial.

As depicted in the latter half of Fig. 2, the process of mixing is illustrated. We utilize P_a and P_b represent the pseudo-labels of Img_a and Img_b, respectively. And the mixing process can be denoted as:

$$MP_{ab} = RM_a * P_a + (1 - RM_a) * RM_b * P_b, \tag{9}$$

$$MP_{ba} = RM_b * P_b + (1 - RM_b) * RM_a * P_a, \tag{10}$$

where the MP_{ab} and MP_{ba} denote two distinct directions of pseudo-label mixing, specifically, $P_a \rightarrow P_b$ and $P_b \rightarrow P_a$, respectively. In this manner, we replace the unreliable pseudo-labels with those that are reliable. Similarly, correspond mixed image MI_{ab} and MI_{ba} are generated as the following:

$$MI_{ab} = RM_a * Img_a + (1 - RM_a) * RM_b * Img_b, \tag{11}$$

$$MI_{ba} = RM_b * Img_b + (1 - RM_b) * RM_a * Img_a. \tag{12}$$

It is noteworthy that both the mixed image and the mixed pseudo-labels might be incomplete, potentially resulting in some hollow areas. The hollow area H denoted as:

$$H_{ab} = 1 - (RM_a + (1 - RM_a) * RM_b), \tag{13}$$

$$H_{ba} = 1 - (RM_b + (1 - RM_b) * RM_a). \tag{14}$$

In such cases, these hollow areas are simply overlooked during the calculation of the loss function. Finally, we amalgamate MI_{ab} and MI_{ba} to formulate a novel training dataset, denoted as MI. Concurrently, we generate corresponding pseudo-labeled supervisory signals, represented as MP and hollow areas as H. We used the same network from the previous step to generate predictions y^{mix} for mixed images. We employ Cross Entropy loss to train the model with the new mixed images. This loss function L_{mix} is defined as:

$$L_{mix} = -\frac{1}{N} \sum_{c=1}^{N} H \cdot \mathbb{I}(MP = n) \cdot \log(H \cdot y_n^{mix}). \tag{15}$$

Indeed, by providing richer supervised signals, we enhance the learning capacity of the model and lead to more accurate and reliable predictions.

3.4. Training with Pick and Mix

Our study underscores the critical role of high-quality, diverse training data in learning processes. We have implemented a two-pronged strategy “Pick” and “Mix” to enhance the quality and quantity of the supervised signals used for training. Note that the training procedure is in an end-to-end manner. The overall training loss can be denoted as:

$$L_{all} = L_{pCE} + \lambda(L_{wp} + L_{mix}), \tag{16}$$

where λ is a weight factor to balance the supervision of scribbles and pseudo labels. We set it as 0.5 in our approach.

4. Experiment

4.1. Datasets

PickMix is evaluated on three public datasets, i.e., ACDC [20], MSCMRseg [21,22] and BraTS2020 [23].

The **ACDC** dataset is comprised of cine-MRI scans from 100 patients. Each patient’s data includes manual scribble annotations for the right ventricle (RV), left ventricle (LV), and myocardium (MYO) provided by Valvano et al. [15]. Following the Zhang and Zhuang [18], these scans are partitioned into three subsets: 70 for training, 15 for validation, and 15 for testing. To facilitate comparison with prior SOTA methods that utilize unpaired masks to learn shape priors, the training set is further divided into two groups: one comprising 35 images with scribble labels and the other consisting of 35 mask images with heart segmentation. It is important to note that, in our approach, the corresponding masks are not utilized during the training process.

The **MSCMRseg** dataset comprises Late Gadolinium Enhancement (LGE) MRI images collected from 45 patients. Each patient’s data includes scribble annotations for the RV, LV, and MYO provided by Zhang and Zhuang [18]. Following the Zhang and Zhuang [18], these scans are partitioned into three subsets: 25 for training, 5 for validation, and 15 for testing.

The BraTS 2020 dataset [23] includes 369 MRI volumes across four modalities (T1, T1ce, T2, and FLAIR), with a uniform resolution of 1.0 mm^3 and a size of $240 \times 240 \times 155$. This study focuses on segmenting the tumor core and peritumoral edema. The dataset was randomly split into 70%, 15%, and 15% for training, validation, and testing, respectively. Scribble annotations were programmatically generated in the axial view to simulate clinician annotations, following Luo [32].

4.2. Implementation details

We employed U-Net [33] as the backbone model, with modifications tailored to the dimensionality of the datasets. Specifically, a 2D U-Net architecture was utilized for the 2D datasets ACDC and MSCMRseg, while a 3D U-Net was adopted for the 3D dataset BraTS2020. During the testing phase, we only employ the standard U-Net model, avoiding any additional computational burden. The models were implemented using the PyTorch framework and trained on a single NVIDIA RTX 3090. For the 2D training datasets, we applied data augmentation techniques such as random rotation, flipping, and the addition of noise, and all 2D images were cropped to a uniform size of 256×256. For the 3D training dataset, data augmentation was limited to random cropping, with a patch size of $80 \times 96 \times 96$ used during training. The batch size was set to 6, and the learning rate was fixed at a constant value of $1e-4$, with a weight decay parameter of 0.0005. The models were trained using the AdamW optimizer over 300 epochs. In our experiments, the Dice coefficient (Dice) and the 95% Hausdorff Distance (HD95) were employed as evaluation metrics.

Table 1
Comparison with different SOTA methods on the ACDC dataset. **Bold** denotes the best performance, underline denotes the second best performance, except for the nnUNet model, which is trained with the full mask.

Method	Data	LV	MYO	RV	Avg
35 scribbles					
UNet_{pce} [12]	Scribbles	0.635	0.563	0.419	0.539
$\text{UNet}_{\text{wpce}}$ [15]	Scribbles	0.784	0.675	0.563	0.674
$\text{UNet}_{\text{ustr}}$ [34]	Scribbles	0.605	0.599	0.655	0.620
$\text{UNet}_{\text{mloss}}$ [35]	Scribbles	0.873	0.812	0.833	0.839
UNet_{em} [36]	Scribbles	0.789	0.761	0.788	0.779
UNet_{crf} [37]	Scribbles	0.766	0.661	0.590	0.672
$\text{UNet}^{+}_{\text{pce}}$ [38]	Scribbles	0.785	0.725	0.746	0.752
$\text{UNet}^{++}_{\text{pce}}$ [39]	Scribbles	0.846	0.787	0.652	0.761
UNETR [40]	Scribbles	0.688	0.330	0.180	0.399
SwinUNETR [41]	Scribbles	0.768	0.683	0.640	0.697
SwinUNet [42]	Scribbles	0.862	0.768	0.735	0.788
TransUNet [3]	Scribbles	0.599	0.475	0.428	0.501
TFCNs [43]	Scribbles	0.703	0.614	0.619	0.645
ScribbleVC [16]	Scribbles	0.914	0.866	0.870	0.884
SC-Net [17]	Scribbles	0.915	0.839	0.862	0.872
VQM [44]	Scribbles	0.919	0.857	0.858	0.881
$\text{nnUNet}_{\text{pL}}$ [45]	Scribbles	0.842	0.843	0.829	0.838
ScribFormer [46]	Scribbles	0.922	<u>0.871</u>	0.871	<u>0.888</u>
ScribSD+ [47]	Scribbles	<u>0.926</u>	0.854	<u>0.873</u>	0.885
Ours	Scribbles	**0.932**	**0.895**	**0.885**	**0.904**
35 scribble + 35 unpaired masks					
UNet_{D} [15]	Mixed	0.404	0.597	0.753	0.585
PostDAE [14]	Mixed	0.806	0.667	0.556	0.676
ACCL [48]	Mixed	0.878	0.797	0.735	0.803
MAAG [15]	Mixed	0.879	0.817	0.752	0.816
35 masks					
UNet_{F} [33]	Masks	0.892	0.830	0.789	0.837
$\text{UNet}^{+}_{\text{F}}$ [38]	Masks	0.849	0.792	0.817	0.820
$\text{UNet}^{++}_{\text{F}}$ [39]	Masks	0.875	0.798	0.771	0.815
$\text{PuzzleMix}_{\text{F}}$ [30]	Masks	0.912	0.842	0.887	0.880
$\text{CycleMix}_{\text{F}}$ [18]	Masks	0.919	0.858	0.882	0.886
UNETR [40]	Masks	0.926	0.844	0.845	0.872
SwinUNet [42]	Masks	0.900	0.812	0.818	0.843
nnUNet [1]	Masks	0.943	0.901	0.915	0.920

Table 2
Comparision with different pseudo-label generating methods on the ACDC dataset. **Bold** denotes the best performance, underline denotes the second best performance.

Methods	Data	LV	MYO	RV	Avg
RandomWalk [49]	Scribbles	0.840	0.730	0.802	0.791
S2L [50]	Scribbles	0.767	0.715	0.765	0.820
SAM [51]	Scribbles	0.911	<u>0.868</u>	0.823	0.868
Ours (only L_{wp})	Scribbles	<u>0.917</u>	0.865	<u>0.846</u>	<u>0.876</u>
Ours (All)	Scribbles	**0.932**	**0.895**	**0.885**	**0.904**

Table 3
Comparison of various state-of-the-art methods on the MSCMRseg dataset. **Bold** indicates the best performance, while underlined text indicates the second-best performance, except for the nnUNet model, which is trained with the full mask.

Methods	Data	LV	MYO	RV	Avg
25 scribbles					
$\text{UNet}^{+}_{\text{pce}}$ [38]	Scribbles	0.494	0.583	0.570	0.378
$\text{UNet}^{++}_{\text{pce}}$ [39]	Scribbles	0.497	0.506	0.472	0.492
ScribFormer [46]	Scribbles	0.896	0.813	0.807	0.839
ScribbleVC [16]	Scribbles	<u>0.921</u>	0.830	0.852	0.868
$\text{nnUNet}_{\text{pL}}$ [45]	Scribbles	0.895	**0.854**	<u>0.870</u>	<u>0.873</u>
Ours	Scribbles	**0.925**	<u>0.837</u>	**0.888**	**0.883**
25masks					
UNet_{F} [33]	Masks	0.850	0.721	0.738	0.770
$\text{UNet}^{+}_{\text{F}}$ [38]	Masks	0.857	0.720	0.689	0.755
$\text{UNet}^{++}_{\text{F}}$ [39]	Masks	0.866	0.745	0.731	0.774
$\text{PuzzleMix}_{\text{F}}$ [30]	Masks	0.867	0.742	0.759	0.789
$\text{CycleMix}_{\text{F}}$ [18]	Masks	0.864	0.785	0.781	0.810
nnUNet_{F} [1]	Masks	0.944	0.882	0.880	0.902

4.3. Comparison with SOTA methods

To comprehensively demonstrate the segmentation performance of our method, we compare our method with various SOTA methods.

- Different scribble-supervised strategies and framework. Comparison with different strategies including partial-cross entropy loss (pce) [12], weight partial-cross entropy loss (wpce) [15], self-ensemble based uncertainty and transformation based consistent (ustr) [34], mumford-shah loss (mloss) [35], entropy minimization (em) [36], and conditional random field (crf) [37]. In addition, comparisons were made with some deformed unet with partial-cross entropy loss, including $\text{UNet}^{+}_{\text{pce}}$ [38], $\text{nnUNet}_{\text{pL}}$ [45] and $\text{UNet}^{++}_{\text{pce}}$ [39].
- Transformer based methods, which combined convolutional layers and Transformers to achieve superior performance in medical image segmentation. Including UNETR [40], SwinUNETR [41], SwinUNet [42], TransUNet [3], TFCNs [43], ScribFormer [46] and ScribbleVC [16].
- Prior knowledge based methods, which incorporate structural or texture-oriented knowledge in the training process. Including SC-Net [17] and VQM [44].
- Additional resources based methods. First, we conduct a comparative analysis with the method that incorporates an unpaired mask. Including UNet_{D} [15], PostDAE [14], ACCL [48] and MAAG [15]. Second, we also make comparison with fully-supervised method, include UNet_{F} [33], $\text{UNet}^{+}_{\text{F}}$ [38], $\text{UNet}^{++}_{\text{F}}$ [39], nnUNet_{F} [1], $\text{PuzzleMix}_{\text{F}}$ [30], $\text{CycleMix}_{\text{F}}$ [18], UNETR [40] and SwinUNet [42].

As demonstrated in Tables 1 and 3, our Pick & Mix model exhibits superior performance over a variety of training strategies, model architectures that are based on UNet in the context of scribble supervision. Additionally, in comparison to the model that employs a combination of Transformer techniques, our approach, which solely utilizes UNet, achieves superior results. Specifically, when compared to the SOTA method, ScribFormer, our proposed method exhibits a performance improvement of 1.6% on the ACDC dataset.

We further compare our method with the model trained with additional unpaired masks. The results of the experiments can be found in the second group of Table 1. It is clear that our method, even without the use of additional resources, has achieved an average performance gain of up to 8.8% over MAAG, which is currently ranked second. We posit that due to the constraints of supervised samples, methods based on Generative Adversarial Networks (GANs) are unable to provide an abundant shape prior for the model. In contrast, our approach selects reliable pseudo labels based on labeled pixels. This strategy offers a more accurate supervision signal compared to unpaired masks, thereby enhancing the effectiveness of our method. Furthermore, the last group of Tables 1 and 3 give the result of several methods trained with fully-supervised mask. Our approach, in particular, exhibits exceptional performance while requiring minimal annotation. This demonstrates the superiority of our method.

4.4. Comparison with pseudo-label generating methods

In this section, we conduct a comparison with several pseudo-label generation methods on the ACDC dataset. We employed a UNet model trained with partial cross-entropy loss as the baseline model and incorporated corresponding methods, including RandomWalk [49], S2L [50], and SAM [51]. Notably, SAM leverages scribble-annotation as a prompt to generate pseudo labels.

As shown in Table 2, when the model is trained using only the reliability of the pseudo-labels as a weight matrix(only the first step of our method, denote as L_{wp} in table), our method demonstrates superior performance compared to SAM, which is the second-best model. Indeed, the utilization of a weight matrix can effectively mitigate

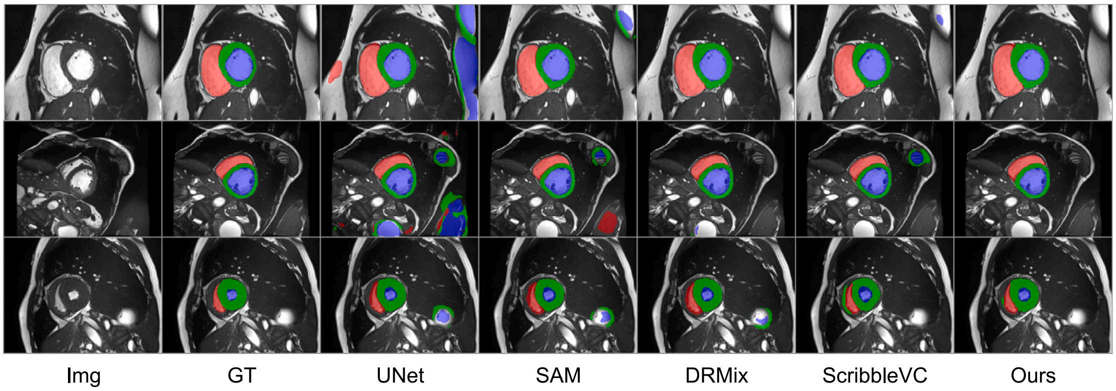

Fig. 4. Quantitative comparison of the proposed method on ACDC.

Table 4
Comparison with different Mixup strategies on the ACDC and MSCMRseg dataset. **Bold** denotes the best performance, underline denotes the second best performance. We referred to their segmentation results reported in [16] for comparison.

Methods	Data	ACDC (35 scribbles)				MSCMRseg (25 scribbles)			
		LV	MYO	RV	Avg	LV	MYO	RV	Avg
MixUp [26]	Scribbles	0.803	0.753	0.767	0.774	0.610	0.463	0.378	0.484
Cutout [27]	Scribbles	0.832	0.754	0.812	0.800	0.459	0.641	0.697	0.599
CutMix [28]	Scribbles	0.641	0.734	0.740	0.705	0.578	0.622	0.761	0.654
Puzzle Mix [30]	Scribbles	0.663	0.650	0.559	0.624	0.061	0.634	0.028	0.241
Co-mixup [29]	Scribbles	0.622	0.621	0.702	0.648	0.356	0.343	0.053	0.251
DRMix [19]	Scribbles	0.875	**0.903**	0.852	0.870	0.881	0.644	0.863	0.796
CycleMix [18]	Scribbles	0.883	0.798	0.863	0.848	0.870	0.739	0.791	0.800
Ours	Scribbles	**0.932**	0.895	**0.885**	**0.904**	**0.925**	**0.837**	**0.888**	**0.883**

Table 5
Comparison with our method and existing weakly supervised methods on 3D BraTS2020. The best values are highlighted in **bold**.

Methods	Tumor		Edema		Mean	
	DSC (%)↑	HD95 (mm)↓	DSC (%)↑	HD95 (mm)↓	DSC (%)↑	HD95 (mm)↓
UNet$_{em}$ [36]	70.98	47.03	65.67	43.73	68.33	45.38
UNet$_{pce}$ [12]	56.76	74.09	50.46	67.87	53.61	70.98
UNet$_{ustr}$ [34]	63.90	69.17	46.88	84.11	55.39	76.64
nnUNet$_{pL}$ [45]	73.63	32.28	63.92	34.04	68.78	33.16
CycleMix [18]	77.67	28.82	64.24	40.58	70.96	34.67
S2L [50]	75.80	17.83	61.50	25.28	68.65	21.55
DMSPS[a] [52]	74.58	12.28	70.78	10.58	72.68	11.43
Ours	**78.01**	**9.71**	**72.77**	**7.50**	**75.39**	**8.61**

[a] Indicate that the first stage of the DMSPS method was retrained in our experimental setup.

the impact of noise information. This, in turn, enables the model to generate reliable and precise predictions, thereby enhancing the overall performance and accuracy of our method. This is a key advantage of our approach. Moreover, our overall approach demonstrates a significant improvement over SAM, with increase of 3.6% on avg dice score. This further validates the effectiveness of our approach.

4.5. Comparison with mix-up strategies

In the second phase of our methodology, we select reliable pseudo labels based on the reliability matrix and mix them to generate a mixed pseudo label as a supervision signal to train the model. To demonstrate the efficacy of this approach, we compare it with several mix-up strategies in this section, including MixUP [26], Cutou [27], CutMix [28], PuzzleMix [30], and Co-mixup [29]. These methods indeed directly utilize mix-up strategies in conjunction with scribble-annotation to augment the supervision signal, which enhances the robustness of the model. Moreover, we have also compared our method with two specialized techniques, namely DRMix [19] and CycleMix [18], which employ mix-up strategies with pseudo labels to provide an enhanced supervision signal.

As outlined in Table 4, our method demonstrates superior performance. In comparison to methods that directly mix scribble-annotations, our approach achieves an improvement of more than 20% in the average Dice scores on the MSCMRseg dataset. Moreover, our method maintains its superior performance even when compared with DRMix and CycleMix. Specifically, we achieve an 8.3% improvement in average Dice scores on the MSCMRseg dataset compared to CycleMix (the second-best method). The primary reasons we attribute the superior performance of our approach to are as follows: (1) The hybrid pseudo-labeling approach can provide more comprehensive supervision compared to the direct hybrid scribble-annotation, thereby significantly enhancing the model. (2) We employ a meticulous selection process for the pseudo-labels to prevent the introduction of noise that could potentially arise from incorporating low-quality pseudo-labels. The high-quality pseudo-labeling mixture introduces a substantial amount of dependable supervised information into the model. This information can be effectively utilized to address the segmentation task based on scribble-supervision. Besides, this approach enhances the model's ability to understand and interpret complex patterns in the data, thereby improving its overall performance in segmentation tasks.

Table 6
Ablation study on the ACDC dataset.

Method	L_{pce}	L_{wp}	L_{mix}	LV	MYO	RV	Avg
#1	✓	✗	✗	0.635	0.563	0.419	0.539
#2	✓	✓	✗	0.917	0.865	0.846	0.876
#3	✓	✗	✓	0.887	0.878	0.872	0.879
#4	✓	✓	✓	0.932	0.895	0.885	0.904

4.6. Result on BraTS2020 dataset

Table 5 shows that our method outperforms existing weakly supervised approaches on the BraTS2020 testing set. For tumor and edema segmentation, our method achieves the highest DSCs (78.01% and 72.77%) and the lowest HD95 values (9.71 mm and 7.50 mm), significantly improving boundary accuracy. Averaged across all categories, our method achieves a mean DSC of 75.39% and a mean HD95 of 8.61 mm, establishing new state-of-the-art performance. These results highlight the robustness and accuracy of our approach in comparison to prior methods.

4.7. Quantitative comparison

Fig. 4 shows the segmentation results obtained by our method and several existing methods: $UNet_{pce}$, SAM, DRMix and ScribbleVC. It is evident that the UNet model generates a significant number of incorrect pseudo-labels. Utilizing these pseudo-labels directly as supervisory signals can potentially mislead the model. In our research, we propose a method of filtering pseudo-labels and integrating reliable ones to provide a richer and more dependable supervise information. Our approach, in comparison to other methods, produces predictions that more closely with the ground truth. This further demonstrates that the use of scribble-based annotation is a promising approach for reducing annotation costs, thereby making it a viable alternative for data annotation in future applications.

4.8. Ablation study

Effectiveness of Components. As shown in Table 6, we employed UNet (#1) equipped with a practical cross-entropy loss as the foundational model, which yielded an average dice score of 53.9%. The incorporation of L_{wp} (#2) as an auxiliary segmentation loss led to a substantial performance enhancement, with the average score increasing by 33.7% to reach 87.6%(#2). Similarly, the exclusive addition of L_{mix} resulted in an average performance surge of 34%, culminating in a score of 87.9%(#3). These findings underscore the efficacy of both L_{wp} and L_{mix} in augmenting the model's performance.

L_{wp} enhances the model's performance by mitigating the noisy information present in the pseudo-labels. L_{mix} bolsters the model's supervisory signals via a hybrid approach. We argue that solely eliminating noise in the pseudo-labels does not introduce additional supervisory signals to the model. And directly integrating pseudo-labels is constrained by their inherent quality, thereby limiting potential performance enhancements.

Our findings advocate for the integration of both methods to furnish the model with high-quality and enriched supervisory signals, thereby catalyzing further performance enhancements. The concurrent application of both as a segmentation loss yields superior scores of 90.4%(#4), with improvements of 2.8% and 2.5% compared to their individual use. This strategy adeptly navigates around the inherent limitations of each method, harnessing their combined strengths to achieve optimal results.

Hyperparameter threshold T analysis. In the secondary phase of our methodology, we employed a fixed threshold to select reliable pseudo-labels for mixup. Additionally, we conducted a sensitivity analysis of the parameter to examine its impact on the ACDC dataset. As illustrated in Fig. 5, the model performance exhibits a substantial improvement when the threshold is increased from 0.5 to 0.7. We attribute this to the inability of a low threshold to effectively filter out the noise information in the pseudo-label, which may mislead the model. And based on the experiment results, we have set the value of T to 0.9 for this study.

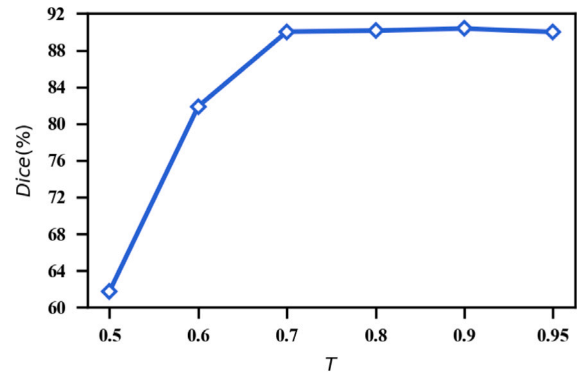

Fig. 5. Sensitivity analysis of hyper-parameter T.

5. Conclusion

In this paper, we propose Pick & Mix, a novel model for scribble-supervised medical image segmentation. The main objective of our model is to provide reliable and rich supervisory signals for the model. To achieve this, our model consists of two main steps. First, we enhance the quality of the supervisory signals by reducing the noise and selecting the most reliable ones for subsequent training. Second, we increase the diversity of the supervisory signals by blending highly reliable ones and generating corresponding blended images. The use of these images for training further augments the segmentation capability of our model. We evaluate the Pick & Mix model on ACDC, MSCMRseg and BraTS2020. The model achieves SOTA performance on these datasets, demonstrating its effectiveness.

CRediT authorship contribution statement

Jiawei Su: Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Zhiming Luo:** Writing – review & editing, Methodology, Funding acquisition, Conceptualization. **Dazhen Lin:** Writing – review & editing, Methodology, Funding acquisition. **Lihui Lin:** Resources, Project administration, Funding acquisition. **Shaozi Li:** Supervision, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

[1] F. Isensee, P.F. Jaeger, S.A. Kohl, J. Petersen, K.H. Maier-Hein, nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation, Nature Methods 18 (2) (2021) 203–211.

[2] Q. Xia, H. Zheng, H. Zou, D. Luo, H. Tang, L. Li, B. Jiang, A comprehensive review of deep learning for medical image segmentation, Neurocomputing (2024) 128740.

[3] J. Chen, Y. Lu, Q. Yu, X. Luo, E. Adeli, Y. Wang, L. Lu, A.L. Yuille, Y. Zhou, Transunet: Transformers make strong encoders for medical image segmentation, 2021, arXiv preprint arXiv:2102.04306.

[4] T. Qu, X. Li, X. Wang, W. Deng, L. Mao, M. He, X. Li, Y. Wang, Z. Liu, L. Zhang, et al., Transformer guided progressive fusion network for 3D pancreas and pancreatic mass segmentation, Med. Image Anal. 86 (2023) 102801.

[5] Z. Hu, H. Chen, L. Hua, X. Ren, W. Mei, MSML-AttUNet: A hierarchical attention network with multi-scale and multi-task for precision liver tumor segmentation, Biomed. Signal Process. Control. 99 (2025) 106861.

[6] Y. Feng, J. Su, J. Zheng, Y. Zheng, X. Zhang, A parallelly contextual convolutional transformer for medical image segmentation, Biomed. Signal Process. Control. 98 (2024) 106674.

[7] W. Shen, Z. Peng, X. Wang, H. Wang, J. Cen, D. Jiang, L. Xie, X. Yang, Q. Tian, A survey on label-efficient deep image segmentation: Bridging the gap between weak supervision and dense prediction, IEEE Trans. Pattern Anal. Mach. Intell. (2023).

[8] B. Rahmati, S. Shirani, Z. Keshavarz-Motamed, Semi-supervised segmentation of medical images focused on the pixels with unreliable predictions, Neurocomputing 610 (2024) 128532.

[9] M. Lin, G. Li, S. Xu, Y. Hao, S. Zhang, Decoupling foreground and background with siamese ViT networks for weakly-supervised semantic segmentation, Neurocomputing 610 (2024) 128540.

[10] M. Li, Y. Gao, W. Cai, W. Yang, Z. Huang, X. Hu, V.C. Leung, Enhanced attention guided teacher-student network for weakly supervised object detection, Neurocomputing (2024) 127910.

[11] C. Liu, Y. Shen, Q. Xiao, G. Li, Pcsformer: Pair-wise cross-scale sub-prototypes mining with CNN-transformers for weakly supervised semantic segmentation, Neurocomputing 593 (2024) 127834.

[12] D. Lin, J. Dai, J. Jia, K. He, J. Sun, Scribblesup: Scribble-supervised convolutional networks for semantic segmentation, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2016, pp. 3159–3167.

[13] I. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, Y. Bengio, Generative adversarial networks, Commun. ACM 63 (11) (2020) 139–144.

[14] A.J. Larrazabal, C. Martínez, B. Glocker, E. Ferrante, Post-DAE: anatomically plausible segmentation via post-processing with denoising autoencoders, IEEE Trans. Med. Imaging 39 (12) (2020) 3813–3820.

[15] G. Valvano, A. Leo, S.A. Tsaftaris, Learning to segment from scribbles using multi-scale adversarial attention gates, IEEE Trans. Med. Imaging 40 (8) (2021) 1990–2001.

[16] Z. Li, Y. Zheng, X. Luo, D. Shan, Q. Hong, Scribblevc: Scribble-supervised medical image segmentation with vision-class embedding, in: Proceedings of the 31st ACM International Conference on Multimedia, 2023.

[17] M. Zhou, Z. Xu, K. Zhou, R.K.-y. Tong, Weakly supervised medical image segmentation via superpixel-guided scribble walking and class-wise contrastive regularization, in: International Conference on Medical Image Computing and Computer-Assisted Intervention, Springer, 2023, pp. 137–147.

[18] K. Zhang, X. Zhuang, Cyclemix: A holistic strategy for medical image segmentation from scribble supervision, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 11656–11665.

[19] X. Luo, M. Hu, W. Liao, S. Zhai, T. Song, G. Wang, S. Zhang, Scribble-supervised medical image segmentation via dual-branch network and dynamically mixed pseudo labels supervision, in: International Conference on Medical Image Computing and Computer-Assisted Intervention, Springer, 2022, pp. 528–538.

[20] O. Bernard, A. Lalande, C. Zotti, F. Cervenansky, X. Yang, P.-A. Heng, I. Cetin, K. Lekadir, O. Camara, M.A.G. Ballester, et al., Deep learning techniques for automatic MRI cardiac multi-structures segmentation and diagnosis: is the problem solved? IEEE Trans. Med. Imaging 37 (11) (2018) 2514–2525.

[21] X. Zhuang, Multivariate mixture model for cardiac segmentation from multi-sequence MRI, in: International Conference on Medical Image Computing and Computer-Assisted Intervention, Springer, 2016, pp. 581–588.

[22] X. Zhuang, Multivariate mixture model for myocardial segmentation combining multi-source images, IEEE Trans. Pattern Anal. Mach. Intell. 41 (12) (2018) 2933–2946.

[23] B.H. Menze, A. Jakab, S. Bauer, J. Kalpathy-Cramer, K. Farahani, J. Kirby, Y. Burren, N. Porz, J. Slotboom, R. Wiest, et al., The multimodal brain tumor image segmentation benchmark (BRATS), IEEE Trans. Med. Imaging 34 (10) (2014) 1993–2024.

[24] Y. Wang, Y. Ji, H. Xiao, A data augmentation method for fully automatic brain tumor segmentation, Comput. Biol. Med. 149 (2022) 106039.

[25] X. Jia, Y. Shen, J. Yang, R. Song, W. Zhang, M.Q.-H. Meng, J.C. Liao, L. Xing, PolypMixNet: Enhancing semi-supervised polyp segmentation with polyp-aware augmentation, Comput. Biol. Med. 170 (2024) 108006.

[26] H. Zhang, M. Cisse, Y.N. Dauphin, D. Lopez-Paz, Mixup: Beyond empirical risk minimization, 2017, arXiv preprint arXiv:1710.09412.

[27] T. DeVries, G.W. Taylor, Improved regularization of convolutional neural networks with cutout, 2017, arXiv preprint arXiv:1708.04552.

[28] S. Yun, D. Han, S.J. Oh, S. Chun, J. Choe, Y. Yoo, Cutmix: Regularization strategy to train strong classifiers with localizable features, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2019, pp. 6023–6032.

[29] J.-H. Kim, W. Choo, H. Jeong, H.O. Song, Co-mixup: Saliency guided joint mixup with supermodular diversity, 2021, arXiv preprint arXiv:2102.03065.

[30] J.-H. Kim, W. Choo, H.O. Song, Puzzle mix: Exploiting saliency and local statistics for optimal mixup, in: International Conference on Machine Learning, PMLR, 2020, pp. 5275–5285.

[31] K. Chaitanya, N. Karani, C.F. Baumgartner, A. Becker, O. Donati, E. Konukoglu, Semi-supervised and task-driven data augmentation, in: Information Processing in Medical Imaging: 26th International Conference, IPMI 2019, Hong Kong, China, June 2–7, 2019, Proceedings 26, Springer, 2019, pp. 29–41.

[32] X. Luo, WSL4MIS, 2021, https://github.com/Luoxd1996/WSL4MIS.

[33] O. Ronneberger, P. Fischer, T. Brox, U-net: Convolutional networks for biomedical image segmentation, in: Medical Image Computing and Computer-Assisted Intervention–MICCAI 2015: 18th International Conference, Munich, Germany, October 5-9, 2015, Proceedings, Part III 18, Springer, 2015, pp. 234–241.

[34] X. Liu, Q. Yuan, Y. Gao, K. He, S. Wang, X. Tang, J. Tang, D. Shen, Weakly supervised segmentation of COVID19 infection with scribble annotation on CT images, Pattern Recognit. 122 (2022) 108341.

[35] B. Kim, J.C. Ye, Mumford–shah loss functional for image segmentation with deep learning, IEEE Trans. Image Process. 29 (2019) 1856–1866.

[36] Y. Grandvalet, Y. Bengio, Semi-supervised learning by entropy minimization, Adv. Neural Inf. Process. Syst. 17 (2004).

[37] S. Zheng, S. Jayasumana, B. Romera-Paredes, V. Vineet, Z. Su, D. Du, C. Huang, P.H. Torr, Conditional random fields as recurrent neural networks, in: Proceedings of the IEEE International Conference on Computer Vision, 2015, pp. 1529–1537.

[38] C.F. Baumgartner, L.M. Koch, M. Pollefeys, E. Konukoglu, An exploration of 2D and 3D deep learning techniques for cardiac mr image segmentation, in: Statistical Atlases and Computational Models of the Heart. ACDC and MMWHS Challenges: 8th International Workshop, STACOM 2017, Held in Conjunction with MICCAI 2017, Quebec City, Canada, September 10-14, 2017, Revised Selected Papers 8, Springer, 2018, pp. 111–119.

[39] Z. Zhou, M.M. Rahman Siddiquee, N. Tajbakhsh, J. Liang, Unet++: A nested u-net architecture for medical image segmentation, in: Deep Learning in Medical Image Analysis and Multimodal Learning for Clinical Decision Support: 4th International Workshop, DLMIA 2018, and 8th International Workshop, ML-CDS 2018, Held in Conjunction with MICCAI 2018, Granada, Spain, September 20, 2018, Proceedings 4, Springer, 2018, pp. 3–11.

[40] A. Hatamizadeh, Y. Tang, V. Nath, D. Yang, A. Myronenko, B. Landman, H.R. Roth, D. Xu, Unetr: Transformers for 3d medical image segmentation, in: Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision, 2022, pp. 574–584.

[41] Y. Tang, D. Yang, W. Li, H.R. Roth, B. Landman, D. Xu, V. Nath, A. Hatamizadeh, Self-supervised pre-training of swin transformers for 3d medical image analysis, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 20730–20740.

[42] H. Cao, Y. Wang, J. Chen, D. Jiang, X. Zhang, Q. Tian, M. Wang, Swin-unet: Unet-like pure transformer for medical image segmentation, in: European Conference on Computer Vision, Springer, 2022, pp. 205–218.

[43] Z. Li, D. Li, C. Xu, W. Wang, Q. Hong, Q. Li, J. Tian, Tfcns: A cnn-transformer hybrid network for medical image segmentation, in: International Conference on Artificial Neural Networks, Springer, 2022, pp. 781–792.

[44] Y. Xu, M. Zhou, Y. Feng, X. Xu, H. Fu, R.S.M. Goh, Y. Liu, Minimal-supervised medical image segmentation via vector quantization memory, in: International Conference on Medical Image Computing and Computer-Assisted Intervention, Springer, 2023, pp. 625–636.

[45] K. Gotkowski, C. Lüth, P.F. Jäger, S. Ziegler, L. Krämer, S. Denner, S. Xiao, N. Disch, K.H. Maier-Hein, F. Isensee, Embarrassingly simple scribble supervision for 3D medical segmentation, 2024, arXiv preprint arXiv:2403.12834.

[46] Z. Li, Y. Zheng, D. Shan, S. Yang, Q. Li, B. Wang, Y. Zhang, Q. Hong, D. Shen, ScribFormer: Transformer makes CNN work better for scribble-based medical image segmentation, IEEE Trans. Med. Imaging (2024).

[47] Y. Qu, T. Lu, S. Zhang, G. Wang, Scribsd+: Scribble-supervised medical image segmentation based on simultaneous multi-scale knowledge distillation and class-wise contrastive regularization, Comput. Med. Imaging Graph. 116 (2024) 102416.

[48] P. Zhang, Y. Zhong, X. Li, ACCL: Adversarial constrained-CNN loss for weakly supervised medical image segmentation, 2020, arXiv preprint arXiv:2005.00328.
[49] L. Grady, Random walks for image segmentation, IEEE Trans. Pattern Anal. Mach. Intell. 28 (11) (2006) 1768–1783.
[50] H. Lee, W.-K. Jeong, Scribble2label: Scribble-supervised cell segmentation via self-generating pseudo-labels with consistency, in: Medical Image Computing and Computer Assisted Intervention–MICCAI 2020: 23rd International Conference, Lima, Peru, October 4–8, 2020, Proceedings, Part I 23, Springer, 2020, pp. 14–23.
[51] A. Kirillov, E. Mintun, N. Ravi, H. Mao, C. Rolland, L. Gustafson, T. Xiao, S. Whitehead, A.C. Berg, W.-Y. Lo, et al., Segment anything, 2023, arXiv preprint arXiv:2304.02643.
[52] M. Han, X. Luo, X. Xie, W. Liao, S. Zhang, T. Song, G. Wang, S. Zhang, DMSPS: Dynamically mixed soft pseudo-label supervision for scribble-supervised medical image segmentation, Med. Image Anal. 97 (2024) 103274.

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